Chat Export for sanidhya.rana_contractor@celanese.com

Assistant: Hello! I'm an AI assistant here to help you narrow down the Material requirements for Grade selection. Could you please tell me which application you're considering?

User: i am considering a radome application

Assistant: To better understand your radome application and suggest suitable materials, could you please clarify:

1. What is the exact use of the polymer in your radome? For example, is it for a consumer device, automotive antenna, telecom base station, or industrial equipment?
2. Where will the radome be used in your solution? Is it for indoor, outdoor, vehicle-mounted, or infrastructure installation? If unsure, examples include: home WiFi router, truck antenna cover, 5G base station, or industrial sensor housing.

Your answers will help me identify the most relevant material requirements for your application.

User: It is a cover for a Sub 6G Antenna

Assistant: Thank you for clarifying. To further refine the material requirements for your Sub 6G Antenna radome cover, could you please provide:

1. What is the operating environment? For example, will the radome be used outdoors (exposed to UV, rain, temperature extremes) or indoors (more stable conditions)?
2. What is the minimum wall thickness of your radome cover?
3. Are there any preferred base materials you have used previously or would like to use for this application?

4. Are you open to alternative material suggestions if there is no strict preference for a specific polymer?

These details will help ensure the material options are well-suited for your application.

User: Yes - it will be outside. I don't have a material preference. I thickness 1mm and Dk has to be lower than 5

Assistant: Here are the key material requirements (CTQs) assumed for your radome cover application:

1. Minimum wall thickness: 1.0 mm
2. Dielectric constant (Dk): Less than 5
3. Low dissipation factor (for RF performance)
4. UV stabilized/weather resistant
5. Low moisture absorption
6. Injection molding process
7. Dimensional stability / low warpage
8. High flow for thin-wall molding
9. Flame rating (UL 94): HB at 1.0 mm (interim assumption)
10. Relative Thermal Index – Electrical (RTI Electrical)

Are all of these requirements correct for your application? Please confirm or let me know if any need adjustment.

User: looks good and i need f1 ,I will consider filler

Assistant: Thank you for confirming the requirements and specifying UL f1 outdoor suitability and openness to fillers. To further refine the material options, could you please clarify:

1. Do you require a specific UL 94 flame rating at 1.0 mm (e.g., V-0, V-2, or is HB acceptable)?
2. Is there a target value for dissipation factor (Df) at the relevant frequency for your antenna?
3. Do you have a preferred filler type (such as glass, mineral, or other), or are you open to suggestions based on performance?

Your answers will help ensure the material suggestions are closely matched to your needs.

User: i dont have preference on these you can recommend the grades based on above discussion

Assistant: Here are potential material options for your radome cover application:

1. FORTRON® 1140L6 (PPS): Offers UL f1 outdoor suitability, Dk < 5, low dissipation factor, high flow for thin-wall molding, and robust weatherability. May be suitable for your application, but RF loss should be validated at sub-6 GHz.
2. FORTRON® 1140L4 (PPS): Similar to above, with slightly different flow characteristics.
3. VECTRA® S471 (LCP): Excellent RF properties (Dk ~4, very low dissipation factor), thin-wall capability, and high flow. UL f1 status needs confirmation; if confirmed, this is a strong option.
4. ZENITE® 251 and ZENITE® 251 ECO-B 152 (LCP): Top RF performance, Dk ~2.7, low moisture, high flow. Formal UL f1 listing is not confirmed; may require additional coating or testing for outdoor compliance.
5. Rynite® 530 NC010 (PET): UL f1, Dk ~3.9, moderate dissipation factor, good

weatherability. RF loss is higher than LCP/PPS, so antenna performance should be validated.

6. Zytel® HTN51G35HSL NC010 (PPA): UL f1, Dk ~4, moderate dissipation factor, improved hydrolysis resistance. RF loss is higher than PPS/LCP.

These grades are potential options based on your requirements and technical data. Please note that final material selection and product design decisions rest with you, and RF performance should be validated in your specific antenna design. Ensure compliance with all applicable standards and regulations. Celanese does not manufacture or certify products for these applications.

Note: This recommendation is generated by an AI assistant and should be considered purely informational.